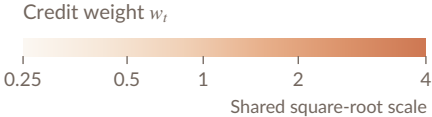


Token-level credit allocation

VPO · Qwen3-14B-Base · $\lambda = 4$



Positive advantage

Deeper orange: stronger reinforcement

(a) Refining an imprecise term

A = +1.47 · rank 1/8

Prompt: What are the lasting legacies of the Greek Empire?

The Greek Empire, or more accurately, Ancient Greece, has left a lasting legacy in various aspects of our world today.

(b) Giving concrete guidance

A = +1.34 · rank 1/8

Prompt: How can a company promote transparency and accountability?

Encourage leaders to model transparency and accountability, and hold them accountable for their actions and decisions.

Negative advantage

Deeper orange: stronger penalty

(c) Using a misleading category

A = -1.94 · rank 8/8

Prompt: What physical factors contribute to the Amazon Rainforest's distinct wildlife?

Tap water: The Amazon Rainforest has a high level of precipitation, ...

(d) Producing an invalid string transformation

A = -2.16 · rank 8/8

Prompt: Remove duplicate characters from QFAWjYIbRIJWvLvKLmdNsW.

... here is the output without duplicate characters:
QFAWjYIKmsNbvm dw

Reference: QFAWjYIbRJvLKmdNs